

Database Management System Course Outline

Course Name: Database Management System (DBMS)

Audience: Students with little to no prior knowledge of DBMS

Duration: 3 hours/day (2:00 pm – 5:00 pm), Monday to Friday

Goal: Enable students to become top 10% database administrators within the country upon completion.

Course Objectives

1. Develop a deep understanding of database concepts, design, and management.
 2. Equip students with practical skills in SQL, database optimization, and performance tuning.
 3. Prepare students to handle database administration in enterprise environments.
 4. Provide knowledge of cutting-edge database technologies and industry practices.
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Week-Wise Course Schedule

Week 1: Introduction to Database Systems

- **Day 1:**
 - What is a Database?
 - Differences between Database and File System.
 - Roles of a Database Administrator (DBA).
 - **Day 2:**
 - Types of Databases (Relational, NoSQL, Object-Oriented).
 - Overview of DBMS Software (MySQL, PostgreSQL, SQL Server, Oracle).
 - **Day 3:**
 - Database Models (Hierarchical, Network, Relational).
 - Components of a DBMS.
 - **Day 4:**
 - Real-Life Applications of Databases.
 - Database Life Cycle.
 - **Day 5:**
 - Introduction to SQL and its Uses.
 - Installing a DBMS (Hands-on: MySQL/SQL Server).
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Week 2: Relational Database Design

- **Day 6:**

- Entities, Attributes, and Relationships.
 - Entity-Relationship (ER) Modeling.
 - **Day 7:**
 - Converting ER Diagrams to Relational Models.
 - Hands-on: Create an ER Diagram for a Sample System.
 - **Day 8:**
 - Relational Algebra and Operations.
 - Hands-on: Write Relational Queries.
 - **Day 9:**
 - Functional Dependencies and Normalization.
 - First to Third Normal Form (1NF, 2NF, 3NF).
 - **Day 10:**
 - Advanced Normalization (BCNF, 4NF, 5NF).
 - Hands-on: Normalize a Sample Database.
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Week 3: SQL and Database Querying

- **Day 11:**
 - SQL Basics: SELECT, INSERT, UPDATE, DELETE.
 - Data Types in SQL.
 - **Day 12:**
 - Filtering and Sorting Data.
 - Aggregate Functions (COUNT, SUM, AVG, MAX, MIN).
 - **Day 13:**
 - Joins (INNER, LEFT, RIGHT, FULL).
 - Subqueries and Nested Queries.
 - **Day 14:**
 - Working with Indexes and Views.
 - Hands-on: Create and Optimize Queries.
 - **Day 15:**
 - Transaction Management.
 - ACID Properties of Transactions.
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Week 4: Advanced Database Administration

- **Day 16:**
 - Introduction to Database Administration Tools.
 - User Management and Security.
 - **Day 17:**
 - Backup and Recovery Strategies.
 - Hands-on: Backup and Restore in MySQL/SQL Server.
 - **Day 18:**
 - Performance Tuning and Query Optimization.
 - Indexing Strategies.
 - **Day 19:**
 - Database Replication and Clustering.
 - Case Study: Implementing Replication in MySQL.
 - **Day 20:**
 - Monitoring and Troubleshooting Databases.
 - Hands-on: Use Monitoring Tools like pgAdmin or SQL Server Profiler.
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Week 5: Emerging Trends and Final Project

- **Day 21:**
 - NoSQL Databases (MongoDB, Cassandra, Redis).
 - When to Use NoSQL vs. Relational Databases.
- **Day 22:**
 - Cloud Databases (AWS RDS, Azure SQL, Google Cloud).
 - Hands-on: Deploy a Database on AWS RDS.
- **Day 23:**
 - Big Data and Database Analytics.
 - Introduction to Hadoop and Spark.
- **Day 24:**
 - Review of Key Concepts and Industry Practices.
 - Prepare for Final Project.
- **Day 25:**
 - Final Project Presentation and Assessment.

Teaching Methods

- **Lectures:** Theory-based sessions to build foundational knowledge.
- **Hands-on Labs:** Practical exercises to reinforce learning.
- **Case Studies:** Real-world examples to connect theory with practice.
- **Industry Tools:** Exposure to tools like pgAdmin, SQL Server Profiler, and AWS RDS.

Final Assessment

- **Project:** Design and implement a fully functional database system for a business scenario (e.g., Library Management, E-Commerce, Hospital Management).
- **Written Exam:** Assess understanding of database design, SQL, and administration.
- **Practical Exam:** Query optimization, backup/restoration, and security configuration.

Outcomes

- Mastery of database concepts and hands-on tools.
- Proficiency in SQL and database management.
- Readiness for entry-level roles as database administrators.
- Solid foundation for advanced certifications (e.g., Oracle DBA, Microsoft SQL Server).